

Design FMEA (qualitative) — VBF-T6R1-DFMEA-01

Qualitative version, based on simulation signals. Ratings: S = severity (high/medium/low), O = occurrence (high/medium/low); basis = magnitude of simulated value vs threshold. RPN = simplified S×O matrix (qualitative caliber).

#	Failure mode	Failure cause	Simulation signal (detectability basis)	S	O	RPN (qual.)	Design-side mitigation
1	Lithium plating at fast charge	end-of-charge anode surface saturation + electrolyte depletion gradients	anode_potential_v min = +10.96 mV at DFN (margin 11.0 mV); SPMe over-predicted margin — fidelity-sensitive signal	high	medium	high-medium	D _e 6.0e-10 m ² /s low-viscosity electrolyte (transport fix, +30.3 mV vs default at DFN); N/P geometry 102.2 μm; r _{neg} 3.5 μm. Charge-protocol headroom recommended
2	Thermal limit exceedance at 4C/45 °C	~6 W DFN heat generation with passive cooling	T _{max} = 321.42 K vs 323.15 K (margin 1.73 K)	high	low	medium	h = 400 W/m ² K active cooling is a design requirement (h=10 fails: 328.9 K, h=200 fails at DFN: 323.80 K)
3	SEI overgrowth	EC solvent reduction at anode; growth diffusion-limited in EC through SEI	sei_thickness_nm_end = 385.3 nm vs 500 nm (margin 114.7 nm)	medium	low	low-medium	EC-lean electrolyte (2270.5 mol/m ³) + EC diffusivity 1.0e-18 m ² /s (L _∞ sqrt(c ₀ ·D _{ec})); real additive package must reproduce effective values
4	Electrolyte oxidative decomposition at 4.7 V	high-voltage window vs electrolyte HOMO level	Not simulated — true DFT endorsement skipped (real_compute=false)	high	unknown	—	LNMO 4.7 V window; recommend HOMO/IE-EA first-principles endorsement of final electrolyte before production
5	Midpoint/plateau sag under load	through-plane electrolyte depletion at ~70% depth	midpoint 4.1145 V vs 4.1 V (margin 14.5 mV); default-D _e DFN measured 4.0974 V (fail)	medium	low	low	D _e transport lever; separator 12 μm retained
6	Capacity shortfall	anode-limited utilization	6.198 Ah simulated vs 4.5 Ah parameter-set nominal (no shortfall)	low	low	low	none
7	Cathode Mn dissolution / high-voltage fade (LNMO-specific)	Mn ³⁺ disproportionation at high voltage	Not simulated — beyond model boundary	medium	unknown	—	literature-known LNMO risk; surface coating recommendation (design note, not simulated)

Conclusion: highest-risk items are #1 (plating — margin 11 mV at DFN, fidelity-sensitive) and #2 (thermal — margin 1.73 K, requires active cooling h=400). Both mitigations are implemented in the shipped design. Items #4/#7 are unquantified (true-compute/experiment boundary) and are flagged for production-line follow-up. Complete FMEA including process/supplier failures: N/A (beyond pure simulation boundary).